MATHEMATICS DEPARTMENT
FAITH · RIGHTEOUSNESS · WISDOM

cognia SCHOOL OF DISTINCTION 2024

مدارس دار الفكر
DAR ALFIKR SCHOOLS

GRADE 11 · ALGEBRA II · TOPIC 6

Lesson 6-7

Geometric Sequences and Series

$$a_n = a_1 \, r^{\,n-1}$$

Journey of 6 stations, 60 minutes

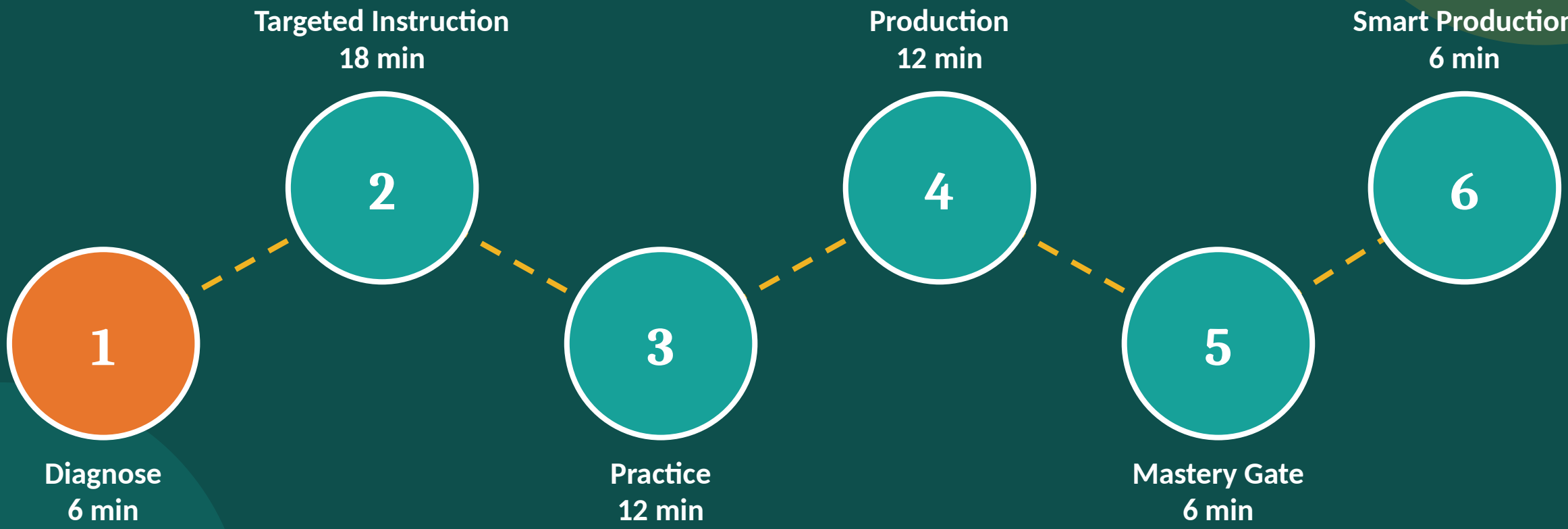
Exponential and Logarithmic Functions

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Today's journey

Six stations. Each one zooms in as we arrive.



What we are learning

OBJECTIVES

1. Identify geometric sequences.
2. Write recursive and explicit formulas for geometric sequences.
3. Identify geometric series.
4. Derive the formula for geometric series.
5. Use the formula for geometric series to find sums.

ESSENTIAL QUESTION

How can we use geometric sequences to model real-world situations?

MATHEMATICAL PRACTICES MP.4 · MP.7

VOCABULARY

- common ratio
- ending index
- geometric sequence
- geometric series
- Index
- sequence of partial sums
- sigma notation
- starting index
- summand
- terms

STANDARDS

HSA.SSE.B.4 Derive the formula for the sum of a finite geometric series (when the common ratio is not 1), and use the formula to solve problems.

HSF.IF.A.3 Recognize that sequences are functions, sometimes defined recursively, whose domain is a subset of the integers.

HSF.BF.A.2 Write arithmetic and geometric sequences both recursively and with an explicit formula, use them to model situations, and translate between the two forms.

HSF.LE.A.2 Construct linear and exponential functions, including arithmetic and geometric sequences, given a graph, a description of a relationship, or two input-output pairs

STATION 1 · DIAGNOSE · 6 MIN

Start here: three quick checks

1

Evaluate $5 \cdot 3^4$

2

Simplify $\dfrac{1 - 2^6}{1 - 2}$

3

Find a_{10}: $7,\ 11,\ 15,\ \ldots$

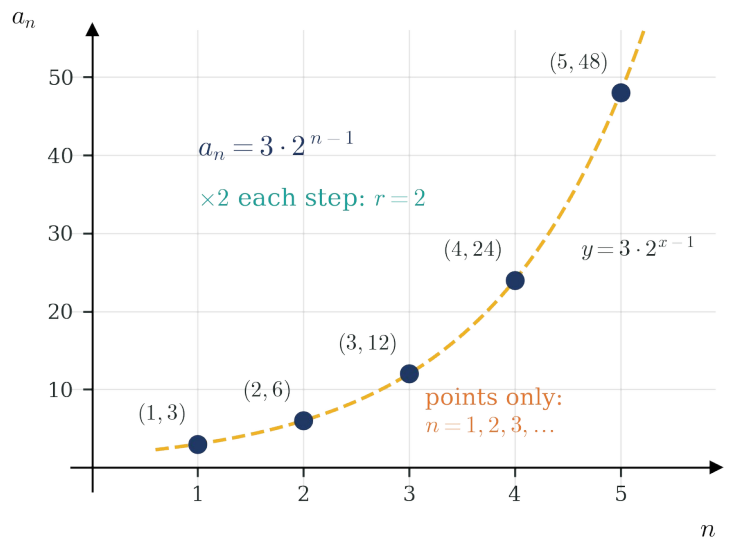
Done when

All three answers are written without a calculator, and you can say which operation repeats in the third list.

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Identify a geometric sequence

$$r = \frac{a_{n+1}}{a_n} \quad \text{(the same for every } n\text{)} \qquad \text{domain: } n = 1, 2, 3, \ldots$$



EXAMPLE A

Is $81,\ -27,\ 9,\ -3$ geometric?

$$\frac{-27}{81} = \frac{9}{-27} = \frac{-3}{9} = -\frac{1}{3}$$

yes: $r = -\frac{1}{3}$

EXAMPLE B

Is $2,\ 4,\ 8,\ 14$ geometric?

$$\frac{4}{2} = 2, \quad \frac{8}{4} = 2, \quad \frac{14}{8} = \frac{7}{4}$$

no: the ratio is not constant

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Recursive and explicit formulas

recursive: a_1 given, $a_n = r\,a_{n-1}$ explicit: $a_n = a_1\,r^{\,n-1}$

EXAMPLE A

$5,\ 15,\ 45,\ \ldots$ Find a_8

$$a_1 = 5,\quad a_n = 3a_{n-1}$$

$$a_n = 5 \cdot 3^{\,n-1}$$

$$a_8 = 5 \cdot 3^{7} = 10\,935$$

EXAMPLE B

$a_2 = 12,\quad a_5 = 96.$ Find a_n

$$r^3 = \frac{96}{12} = 8 \Rightarrow r = 2$$

$$a_1 = \frac{12}{2} = 6$$

$$a_n = 6 \cdot 2^{\,n-1}$$

EXAMPLE C

$$a_1 = 64,\quad a_n = \tfrac{1}{2}a_{n-1}$$

Find a_7

$$a_n = 64\left(\tfrac{1}{2}\right)^{n-1}$$

$$a_7 = 64\left(\tfrac{1}{2}\right)^{6} = \frac{64}{64} = 1$$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Geometric series, sigma notation and the sum formula

$$S_n = \sum_{k=1}^{n} a_1 r^{k-1} \qquad k:\ 1 \to n,\quad \text{summand } a_1 r^{k-1}$$

EXAMPLE A

Derive $S_n \quad (r \neq 1)$

$$rS_n = a_1 r + a_1 r^2 + \cdots + a_1 r^n$$

$$S_n - rS_n = a_1 - a_1 r^n$$

$$S_n = \frac{a_1 (1 - r^n)}{1 - r}$$

EXAMPLE B

Evaluate $\sum_{k=1}^{4} 3 \cdot 2^{k-1}$

$$3 + 6 + 12 + 24$$

$$= 45$$

EXAMPLE C

Partial sums of $2 + 6 + 18 + 54$

$$S_1 = 2,\ S_2 = 8,\ S_3 = 26,\ S_4 = 80$$

check: $\dfrac{2\,(1 - 3^4)}{1 - 3} = \dfrac{-160}{-2} = 80$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Use the formula to find sums

$$S_n = \frac{a_1\,(1 - r^{\,n})}{1 - r}, \quad r \neq 1 \qquad r = 1: \quad S_n = n\,a_1$$

EXAMPLE A

Evaluate $\sum_{k=1}^{8} 5 \cdot 3^{\,k-1}$

$$a_1 = 5,\ r = 3,\ n = 8$$

$$S_8 = \frac{5\,(1 - 3^8)}{1 - 3} = \frac{5(-6560)}{-2}$$

$$S_8 = 16\,400$$

EXAMPLE B

$2 - 6 + 18 - \cdots$: find S_6

$$a_1 = 2,\ r = -3,\ n = 6$$

$$S_6 = \frac{2\,(1 - (-3)^6)}{1 - (-3)} = \frac{2(-728)}{4}$$

$$S_6 = -364$$

EXAMPLE C

$S_n = 765$ for $3 + 6 + 12 + \cdots$. Find n

$$\frac{3\,(1 - 2^{\,n})}{1 - 2} = 765$$

$$2^n - 1 = 255 \Rightarrow 2^n = 256$$

$$n = 8$$

STATION 3 · PRACTICE · 12 MIN

Give it a go: Practice, Apply, Investigate

PRACTICE
Secure the method

$4,\ 12,\ 36,\ \ldots$: find r, a_6

$80,\ 40,\ 20,\ \ldots$: write a_n

Done when: the ratio is found by dividing consecutive terms, and each formula reproduces the first three terms when you substitute the term numbers 1, 2 and 3.

APPLY
Use it in context

$a_3 = 18,\ a_6 = 486$: find $a_1,\ r$

Evaluate $\sum_{k=1}^{6} 2 \cdot 3^{k-1}$

Done when: the number of steps between the two given terms is shown, and the sum names the first term, the ratio and the number of terms before the formula is used.

INVESTIGATE
Find out why

S_n for $5 + 5 + 5 + \cdots \quad (r = 1)$

S_{10} for $1 + \frac{1}{2} + \frac{1}{4} + \cdots$

Done when: you explain why the sum formula fails when the ratio is 1, and you say what value the partial sums of the second series approach, supported by at least three partial sums.

STATION 4 · PRODUCTION · 12 MIN

Production: Jeddah greening programme

THE MODEL

Vision 2030 sets targets for greening Saudi cities. A model Jeddah neighbourhood programme (model values only, not real data) plants 256 trees in month 1, and each month it plants 1.5 times as many trees as the month before. Each sapling costs a model 25 SAR.

$$a_1 = 256, \quad r = 1.5 \quad (\text{trees planted in month } n)$$

$$\text{cost: } 25 \text{ SAR per sapling}$$

1. Write the recursive and explicit formulas, and find the number of trees planted in month 6.
2. Use the sum formula to find the total planted in the first 6 months and its cost in SAR.
3. Find the first month in which the running total passes 10 000 trees; check with the totals for the month before and that month.

DONE WHEN

- Both formulas are written, and month 6 gives 1 944 trees.
- The 6-month total is 5 320 trees, checked by adding the six terms.
- The cost is 133 000 SAR, written with units.
- Month 8 is justified: the total is 8 236 after month 7 and 12 610 after month 8.

Hint for solving: name the first term, the ratio and the number of months first; for Task 3 make a table of running totals, or set the sum formula greater than 10 000.

STATION 5 · MASTERY GATE · 6 MIN

Mastery Gate: three questions to pass

1

Next term: $96,\ -48,\ 24,\ \ldots$

2

$a_1 = 7,\ a_n = 2a_{n-1}$: find a_6

3

Evaluate $\sum_{k=1}^{5} 4 \cdot 3^{k-1}$

Pass mark

3 of 3 correct unlocks Station 6. Fewer than that: return to the slide for the question you missed, redo it, then re-take the gate.

STATION 6 · SMART PRODUCTION · 6 MIN

Smart Production: make one, then check it

CREATE

1. Invent a geometric sequence with a real context: give the first term, the ratio and what the term number counts.
2. Write its recursive and explicit formulas and the sum of its first 5 terms.
3. Swap with a partner: check their sum by adding the five terms.

Done when: your sequence has a constant ratio that is not 1, both formulas give the same first three terms, and your partner's addition matches your formula.

TIME TO CHECK

Exit ticket: write your answer on a sticky note and place it on the door.

$$\text{Find } S_6 \text{ for } 3 + 6 + 12 + \cdots$$

$$a_1 = 1000,\ a_n = 0.8\,a_{n-1}\text{: find } a_5$$

Next lesson: 9-1 Parabolas.

Exam connections

SAT

The first term of a sequence is 2, and each term after the first is 3 times the preceding term.

What is the sum of the first 5 terms?

STEPS

1 $$S_5 = \frac{2\left(3^5 - 1\right)}{3 - 1}$$

2 $$= 3^5 - 1$$

3 $$= 242$$

GAT

Which quantity is greater?

A: 6th term of $1,\ 2,\ 4,\ \ldots$
B: sum of its first 5 terms

STEPS

1 $$\text{A} = 2^5 = 32$$

2 $$\text{B} = 1 + 2 + 4 + 8 + 16 = 31$$

3 $$32 > 31 \Rightarrow \text{A is greater}$$

SAAT

$a_3 = 12,\quad a_6 = 96$ (geometric)

Find S_5

STEPS

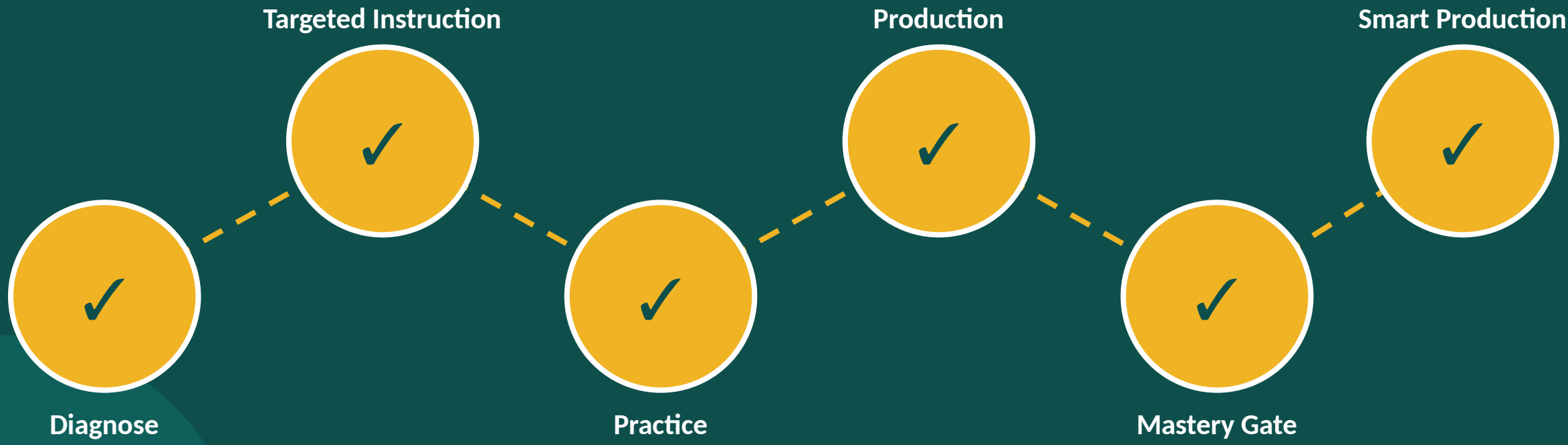
1 $$r^3 = \frac{96}{12} = 8 \Rightarrow r = 2$$

2 $$a_1 = \frac{12}{2^2} = 3$$

3 $$S_5 = 3\left(2^5 - 1\right) = 93$$

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Journey complete



A geometric sequence multiplies by a constant ratio; a geometric series adds its terms, and when the ratio is not 1 the sum has a closed formula.